SYNAPTIX

Clinical-Grade Sleep Labs — Anywhere.

AI-powered Level II home sleep diagnostics platform.

Building the infrastructure for Sleep Intelligence.

ECG + EEG Multi-Modal

Proprietary Time-Series AI

Hardware-Software Co-Design

70M+

Americans with Sleep Disorders

$3B

U.S. TAM
(5M tests × $600)

80%

Reduction in Scoring Time

EXECUTIVE SUMMARY

The Problem

70M+ Americans suffer from sleep disorders. In-lab PSG costs $2,000–$5,000 with weeks-to-months wait times. Consumer wearables lack clinical accuracy. The result: a massive, dangerous diagnostic gap.

Our Solution

Synaptix is an end-to-end, AI-powered, clinical-grade home sleep platform: ECG+EEG wearable sensing hub + intelligent mobile app + cloud-based time-series AI. Clinical accuracy. Home comfort.

Go-to-Market: Land & Expand

Phase 1: AI-Scoring SaaS for existing sleep clinics — eliminate manual scoring backlogs today. Phase 2: Deploy our proprietary Type-II HST device into those same trusted labs.

Market Opportunity

U.S. TAM: $3B (5M annual sleep tests × $600 blended ASP). SAM: $600M. Home sleep testing is the fastest-growing segment. 80% of OSA cases remain undiagnosed.

Core Innovation

Hardware-software co-designed time-series AI. Multi-modal ECG+EEG fusion for 85%+ scoring agreement. Proprietary architecture that gets smarter with every study via data flywheel.

The Ask

Raising a Seed Round. 18-month runway to clinical validation, FDA 510(k) submission, and first recurring SaaS revenue. Capital allocation: 40% R&D, 25% Clinical, 20% Regulatory, 15% Mfg.

THE PROBLEM

Sleep disorders are a silent epidemic — and the diagnostic system is broken.

70M+
Americans with Sleep Disorders

$4,000
Average In-Lab PSG Cost

12 Wks
Average Wait for Diagnosis

80%
OSA Cases Still Undiagnosed

Too Expensive

Gold-standard PSG costs $2,000–$5,000 per study — prohibitive for most patients and strained payer systems.

Too Slow

6–12 week wait times at sleep labs mean delayed diagnoses, worsening conditions, and frustrated patients.

Consumer Wearables Fall Short

Oura, Apple Watch, and Fitbit use actigraphy — not diagnostic-grade biosignals. Clinicians cannot act on their outputs.

THE SYNAPTIX PLATFORM

An end-to-end, clinical-grade sleep diagnostic ecosystem

📡 Edge Sensing Hub

Hardware

- → Bluetooth SoC — low-power, reliable
- → Multi-lead ECG acquisition
- → EEG signal acquisition
- → 12+ hour battery for overnight use
- → Designed for overnight comfort

📱 Intelligent Mobile App

Software

- → Real-time signal validation
- → Guided patient setup flow
- → Secure end-to-end data upload
- → User-friendly results reporting
- → Provider notification system

🧠 AI Cloud Engine

AI / Cloud

- → Time-series deep learning models
- → Multi-modal ECG+EEG signal fusion
- → Sleep stage classification (5-stage)
- → Apnea & arrhythmia detection
- → Longitudinal sleep analytics

Powered by hardware-software co-design: signal preprocessing optimized at the hardware layer — reducing cloud cost and improving AI accuracy.

THE COMPETITIVE GAP

There is no scalable, clinical-grade home solution. Until now.

	In-Lab PSG	Consumer Wearables (Oura, Apple, Fitbit)	Synaptix
Accuracy	✓ High	✗ Low (actigraphy)	✓ Clinical-Grade
Cost	✗ $2,000–$5,000	~ $300–$500	✓ Affordable
Wait Time	✗ 6–12 weeks	✓ Immediate	✓ 24–48 hours
Setting	✗ Hospital/Clinic	✓ Home	✓ Home
Analytics	Manual scoring	Black-box actigraphy	ECG+EEG AI — Automated
Diagnostic Grade	✓ Yes	✗ No	✓ Yes
Scalable	✗ No	✓ Yes	✓ Yes

Synaptix is the only solution that delivers clinical-grade accuracy at home, at scale.

CORE INNOVATION — TIME-SERIES AI

Hardware-Software Co-Designed Time-Series AI

Signal Pre-Processing at Hardware

Preprocessing optimized at the chip level reduces noise before transmission — enabling higher-fidelity data and lower cloud compute cost.

Feature Extraction Co-Design

Model architecture and feature engineering are co-designed together — yielding significantly better classification performance vs. off-the-shelf approaches.

Adaptive Noise Handling

Real-world home environments introduce signal artifacts. Our adaptive models are specifically trained to handle dropout, motion, and interference.

Efficient Inference Pipeline

Inference engine is optimized for low-latency cloud delivery — enabling 48-hour turnaround and cost structure compatible with $25–$75/study SaaS pricing.

The "Eureka" Insight

Sleep physiology is fundamentally a multi-modal time-series problem.

- ECG-derived autonomic features enhance sleep staging accuracy
- EEG+ECG fusion improves apnea detection vs. single-signal methods
- Longitudinal model predicts sleep deterioration trends over time

Target validation metrics:

· ≥ 85% agreement with manual scoring (Cohen's kappa)

· Clinically meaningful apnea detection sensitivity/specificity

· Stable performance under real-world noise and signal dropout

GO-TO-MARKET: LAND & EXPAND

A two-phase strategy that builds trust before deploying hardware.

PHASE 1 — LAND

AI-Scoring SaaS for Sleep Clinics

Solve the immediate pain: eliminate the manual scoring backlogs that plague sleep labs today

Deploy lightweight cloud AI that connects seamlessly with existing lab workflows and EMR systems

Establish trust by proving clinical-grade accuracy to sleep physicians and lab directors

Build a sticky, recurring revenue digital footprint across a growing network of partner labs

Revenue model: $25–$75 per study or per-patient SaaS subscription

PHASE 2 — EXPAND

Type-II HST Device into Trusted Labs

Leverage established SaaS footprint to introduce proprietary Type-II Home Sleep Test device

Deploy directly into trusted, partner labs — reducing friction and bypassing cold-start problem

Unlock CPT 95800 / 95806 reimbursement: $300–$450 per study. Hardware ASP: $600–$900/device

Consumables revenue: <$25 disposable electrode patches per study

Blended margin improves significantly as SaaS + hardware revenues compound together

MARKET OPPORTUNITY

TAM

$3B

U.S. Total Addressable Market

SAM

$600M

Sleep Clinics + Telehealth

SOM

$18M

Year 1–2 Reachable

TAM Build-Up

5M annual
U.S. sleep tests

× $600 blended revenue/test

= $3B annual TAM

SAM ~1M tests
= $600M

- 80% of OSA cases remain undiagnosed
- Level 2 "one-and-done" replaces $2,000+ in-lab PSG
- Clinics moving to Level 2-first models
- Telehealth & RPM expansion driving HSAT adoption
- CPT 95800/95806 reimbursement fully established

SYNAPTIX UNFAIR ADVANTAGES

Hard to replicate without: systems engineering expertise + AI specialization + medical device know-how

Multi-Modal ECG + EEG

Most home sleep tests capture airflow only. Our dual-biosignal approach captures both cardiac and brain activity — dramatically improving diagnostic completeness.

Proprietary Time-Series AI

Custom deep learning architecture built specifically for physiological time-series. Not a wrapper around generic models — trained from the ground up on sleep data.

Hardware-Software Co-Design

Signal preprocessing is optimized at the hardware layer in co-design with the AI model architecture. This yields better accuracy and lower cloud inference cost.

Clinical Validation Pathway

Operating on the FDA 510(k) predicate pathway with AASM-compliant data collection. We're building clinical evidence, not just consumer data.

Dual Hardware + SaaS Model

SaaS revenue starts before hardware launch — creating immediate recurring ARR while building the trusted network into which we expand our device.

Data Flywheel

Every study improves our AI models. As the network of partner labs grows, model accuracy compounds — creating a moat that widens over time against new entrants.

BUSINESS MODEL & FINANCIAL PROJECTIONS

Hardware

Reusable device ASP
$600–$900

SaaS Analytics

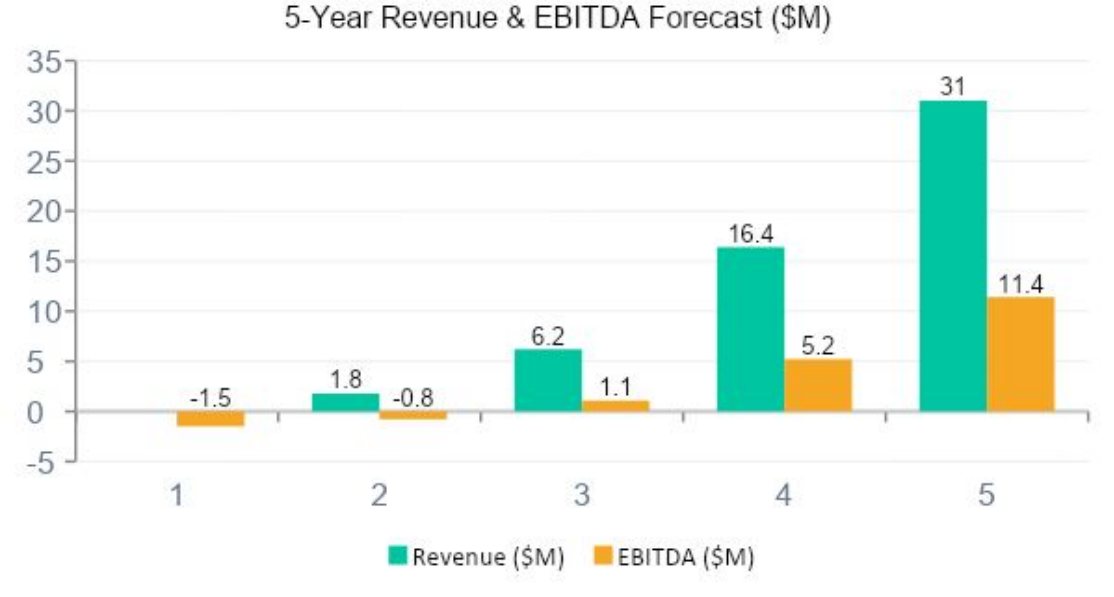
$25–$75 / study
or per-patient/month

Consumables

Electrode patches
<$25 / study

Blended Margin

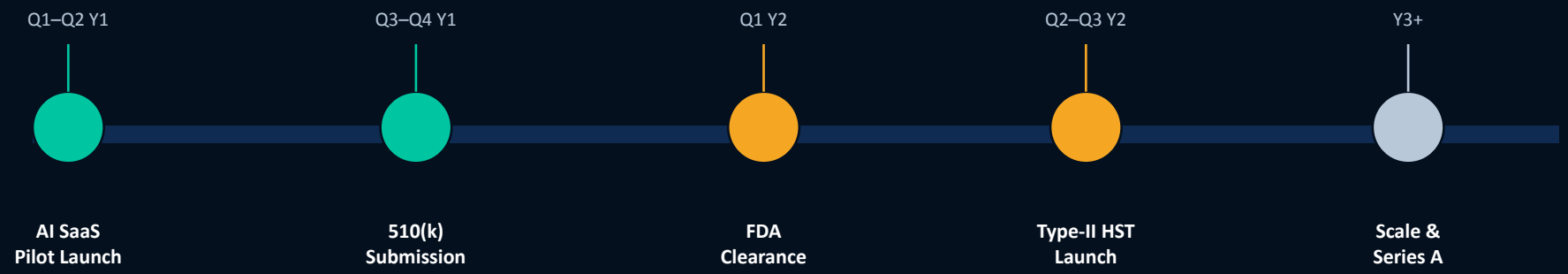
Improves as SaaS
revenue scales



Metric	2025	2026	2027	2028
Studies	0	3,000	10,300	27,300
Revenue	$0	$1.8M	$6.2M	$16.4M
Gross Margin	—	52%	65%	71%
EBITDA	-$1.5M	-$0.8M	$1.1M	$5.2M
Lab Partners	5	150	500	1,200

Key assumptions: $600 blended ASP (SaaS $50 avg + hardware amortized). COGS reduces from $290 → $170/study at scale. 85% annual SaaS retention. 40% YoY lab partner growth.

EXECUTION ROADMAP



AI SAAS LAUNCH

Q1–Q4 Year 1

- Expanded dataset collection & AI model training
- Pilot home study trials with partner clinics
- Clinical-grade validation study (n=150+)
- FDA 510(k) submission — predicate: existing Level II HST
- Sleep clinic and telehealth provider partnerships begin

HARDWARE LAUNCH

Year 2

- FDA 510(k) clearance received
- Type-II HST device deployed to SaaS partner labs
- SaaS commercialization at scale — 150+ labs
- CPT 95800/95806 reimbursement activation
- Insurance and RPM partnership negotiations

SCALE & EXPAND

Year 3+

- Scale manufacturing; reduce COGS to target margin
- Insurable reimbursement pathways nationwide
- Remote Patient Monitoring (RPM) platform expansion
- Series A: $10–15M post FDA clearance + revenue
- Long-term: cardiometabolic AI monitoring platform

LEADERSHIP TEAM

AM

Abdul Mohsin

Co-Founder

Leads product vision and business strategy. Deep expertise in medical device commercialization and health-tech startup development.

TK

Trevor Krawchuk

Co-Founder

Co-founder driving engineering and operations. Background in embedded systems and biomedical device architecture.

ML

Ming Liu

Co-Founder / Advisor

CS Faculty at UW-Madison. Specialization in Networking and Systems. Provides academic research and clinical computing expertise.

CORE TEAM

Q

Qianliang Wu

Founding Engineer · Core platform architecture and embedded systems

B

Ben Levy

Embedded Engineer · Hardware firmware and sensor integration

K

Krish

Engineering · Joining the core technical team

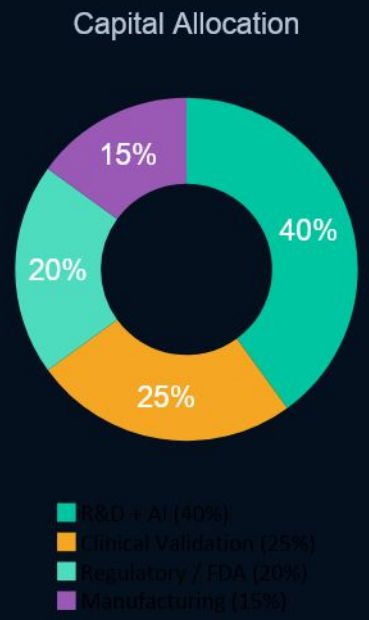
M

Matt Thiel

Engineering · Core team member

THE ASK — SEED ROUND

18-Month Runway to Clinical Validation, FDA Submission & First Recurring ARR



M1–3	Expand dataset collection; begin AI model v2 training; initiate clinical partnerships
M4–6	Pilot SaaS launch with 5–10 sleep clinic partners; real-world validation begins
M7–9	Clinical validation study complete (n=150+); FDA 510(k) pre-submission meeting
M10–12	FDA 510(k) submission filed; SaaS at 50+ lab partners; initial ARR established
M13–15	FDA clearance received; Type-II HST device soft launch into partner labs
M16–18	Scaled commercial launch; Series A preparation; reimbursement pathway activation

Future Series A: Post-FDA clearance — scale sales & manufacturing. | 24-month milestones: clinical validation data, regulatory submission, pilot customers, initial ARR.

SYNAPTIX

Building the Infrastructure for Sleep Intelligence

5-Year Vision

- Leading provider of medical-grade home sleep diagnostics
- Integrated into telehealth & RPM workflows nationwide
- Profitable SaaS-driven health platform
- $31M+ ARR with 1,200+ lab partners

10-Year Vision

- Continuous physiological AI monitoring platform
- Predictive cardiometabolic risk detection at scale
- Global digital health infrastructure company
- Expansion into heart failure, COPD, and beyond

Contact: info@synaptix.ai | www.synaptix.ai